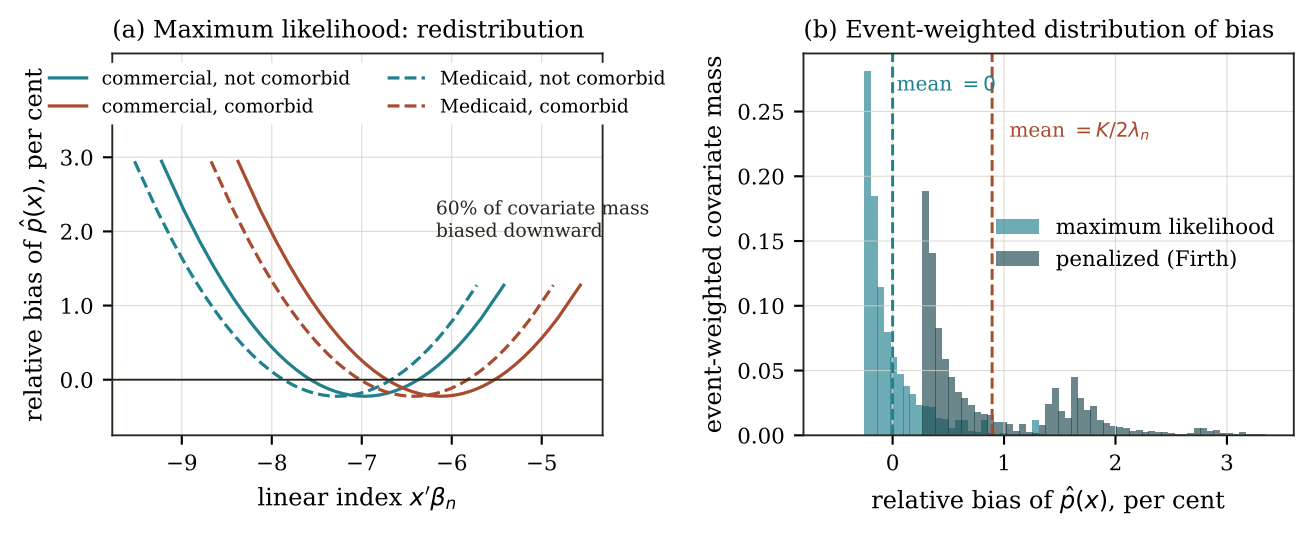
(a) Maximum likelihood: redistribution
commercial, not comorbid
commercial, comorbid
Medicaid, not comorbid
Medicaid, comorbid
60% of covariate mass
biased downward
relative bias of $\hat{p}(x)$, per cent
3.0
2.0
1.0
0.0
−9
−8
−7
−6
−5
linear index $x'\beta_n$
(b) Event-weighted distribution of bias
mean = 0
mean = $K/2\lambda_n$
maximum likelihood
penalized (Firth)
event-weighted covariate mass
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3
relative bias of $\hat{p}(x)$, per cent